

Expansive Field-of-View Head-Mounted Display based on Dynamic Projection Mapping

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ABSTRACT

In this study, we propose a method for extending the peripheral field of view of conventional head-mounted displays (HMDs) based on dynamic projection mapping. External projectors are used to dynamically project images onto peripheral screens attached to the periphery of the HMD. A wide field of view is achieved without significantly increasing the HMD weight.

CCS CONCEPTS

• **Human-centered computing** → **Displays and imagers; Virtual reality.**

KEYWORDS

Head-Mounted Display (HMD), Peripheral Screen, Dynamic Projection Mapping, Delay Compensation, Wide Field of View

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1 INTRODUCTION

Head-mounted displays (HMDs) are used to implement virtual reality (VR) experiences. However, they have a limited field of view, which significantly diminishes the sense of immersion. The development of HMDs capable of achieving an ultra-wide field of view that approaches that of human vision has thus been a focus of research. Electronic components such as small displays [Nakano et al. 2021] and LED arrays [Robert Xiao, Hrvoje Benko 2016] have been integrated within the HMD to extend the field of view. However, these approaches increase the complexity and weight of the device and thus increase user discomfort.

Therefore, in this study, we propose a method that extends the peripheral field of view of conventional HMDs by dynamically projecting images onto lightweight screens attached to the periphery of the HMD.

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2 PROPOSED APPROACH

We achieve a wide field of view by utilizing lightweight screens that cover a broad region of the peripheral field of view not covered by conventional HMDs. This is accomplished by projecting images onto these screens via external projectors. The images displayed on the HMD are combined with the images projected onto the screens. The configuration of the proposed system is shown in Figure 1.

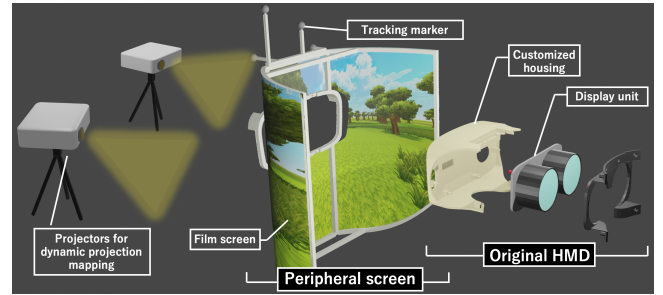


Figure 1: System overview

In the proposed method, the high-resolution images from the HMD cover the central visual field, where perceptual acuity is high, and the relatively low-resolution images from the projector cover the peripheral visual field, where perceptual acuity is lower. Users perceive detailed stereoscopic images in the central visual field and sufficiently accurate images over a wide range of the peripheral visual field. By providing ample information for motion perception through optical flow in the peripheral visual field, which is not achieved by conventional HMDs, the proposed system enables a wide field of view and thus a highly immersive visual experience.

2.1 Design of Peripheral Screen

The design of the peripheral screen is shown in Figure 2. The shape of the screen is designed to efficiently cover the user's peripheral field of view, following a curved surface along the horizontal direction to conform to the head. To facilitate movement of the head during wear, the lower corners of the screen, which are outside the user's field of view, are rounded. A frame with this shape that holds a film screen is used as a lightweight screen that covers a wide field of view.

When images are displayed using this screen in the peripheral field of view, the maximum horizontal field of view is extended up to 226 degrees, which is more than 85% wider than that for the conventional HMD without the screen.

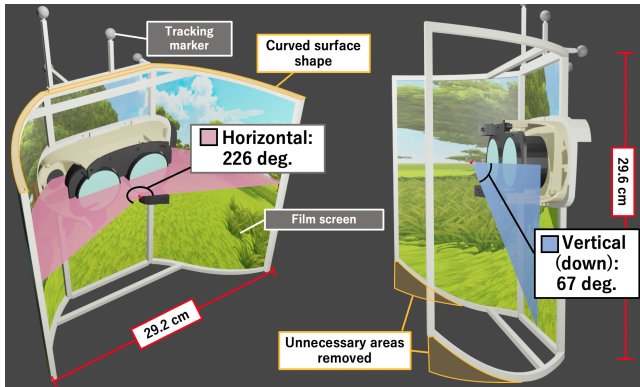


Figure 2: Design of peripheral screen (FOV: field of view)

2.2 Projection Mapping on Peripheral Screen

In the proposed system, dynamic projection mapping is used to project images onto the designed screen. Projected images are created by applying geometric correction using two-pass rendering to normal rendering images. By executing these image correction processes every frame, it is possible to track the moving screen using optical motion capture and perform projection mapping.

Moreover, to minimize the delay of projected images, a predictive method based on the recursive least squares method is utilized to reduce projected image misalignment.

3 IMPLEMENTATION

The configuration of the peripheral section is shown in Figure 1. The total weight of the device is 604 g, approximately 8 g lighter than the original HMD (Dell Visor). This reduction in weight is attributed to the redesign of the HMD housing to allow observation of the peripheral screen and the lightweight design of the screen.

The projection system consists of two projectors (ViewSonic PX701-4K) capable of projecting at a maximum refresh rate of 240 Hz in Full HD and six motion capture cameras (OpiTrack PrimeX 13) capable of capturing images at 240 fps. In the experimental setup, these devices were used to construct a projection area that resembled a right-angled isosceles triangle with equal sides of approximately 2.6 m in length. The projected images could be presented with a pixel density of 1.0 pppd or higher.

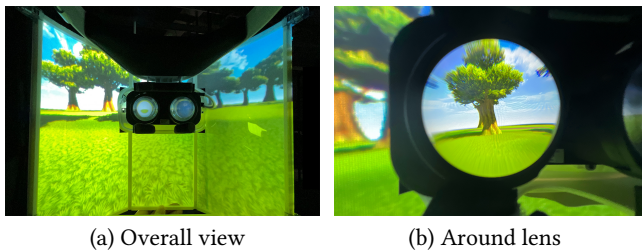


Figure 3: Extended peripheral area of HMD

In the experimental environment, the proposed system projected images onto the entire screen, as shown in Figure 3(a). As shown in

Figure 3(b), the user perceived stereoscopic images from the HMD in the central visual field and projected images on the screen in the peripheral visual field, resulting in the perception of a wide field of view.

This configuration using two projectors is the minimal setup. By surrounding the participant with more projectors, it is possible to expand the range of movement for the participant and present higher resolution images for the peripheral visual field.

4 EVALUATION

To confirm the effectiveness of the proposed system, an experiment with 13 participants (aged 22 to 27) was conducted. In this experiment, participants experienced immersive VR scenes using the HMD with and without the proposed system. After each experience, participants provided a subjective rating of the experienced content and the usability of the proposed system. The Visual Analog Scale was used for evaluation, where each item was rated on a scale of 0 to 10.

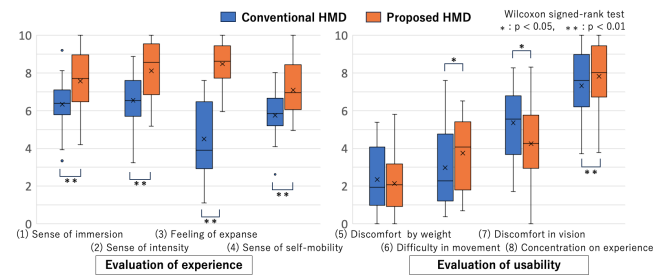


Figure 4: Evaluation of VR experiences

The experimental results are shown in Figure 4. As can be seen, the evaluation scores for the proposed system were significantly higher than those for the conventional HMD in almost all categories. In comments, the participants frequently mentioned that the wide field of view enabled by the proposed system allowed them to feel the images throughout their entire field of view, resulting in a strong sensation of actually being present in the environment and feeling their own movement.

Although there was no significant difference in discomfort due to weight, the discomfort caused by restricted movement was significantly higher for the proposed HMD than for the conventional HMD. This suggests that the peripheral screen may have made contact with the body, causing discomfort. This discomfort could be alleviated by modifying the design of the screen.

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